

Dr. Santiago Rubén Soler

Postdoctoral Research Fellow

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Professional Appointments

- | | |
|-----------|---|
| 2022–on | Postdoctoral Research Fellow
Department of Earth, Ocean and Atmospheric Sciences
University of British Columbia |
| 2017–2019 | Assistant Professor of Practice
Facultad de Ciencias Exactas, Físicas y Naturales,
Universidad Nacional de San Juan, San Juan, Argentina |
| 2013–2015 | Student Teacher
Facultad de Ciencias Exactas, Ingeniería y Agrimensura,
Universidad Nacional de Rosario, Rosario, Argentina |

Education

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|-----------|--|
| 2017–2022 | PhD in Geophysics
Facultad de Ciencias Exactas, Físicas y Naturales,
Universidad Nacional de San Juan, San Juan, Argentina |
| 2009–2015 | Licentiate in Physics
Facultad de Ciencias Exactas, Ingeniería y Agrimensura,
Universidad Nacional de Rosario, Rosario, Argentina |

Certifications

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|------|---|
| 2021 | Certified Software Carpentry Instructor |
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Open-source and Community Involvement

- Since 2015 **Fatiando a Terra: Open source tools for geophysics**
Maintainer and Project Leader
<https://www.fatiando.org>
- Since 2023 **SimPEG: Simulation and Parameter Estimation in Geophysics**
Maintainer
<https://simpeg.xyz>
- Since 2023 **Software Underground**
Board member
<https://softwareunderground.org>

Awards and Scholarships

- 2017–2022 PhD Scholarship from CONICET
- 2019 Early Career Scientist's Travel Support for EGU2020 General Assembly
- 2016 Fundación Josefin Prats Award: Licenciatura en Física
- 2012 University Scholarship granted by Fundación del Banco de Santa Fe
- 2012 Santander Río National Award to Academic Merit 2012
- 2012 Josefin Prats Award: IFIR 2012

Teaching Experience

UNDERGRADUATE COURSES

- 2022–2026 **EOSC 350: Environmental, Geotechnical, and Exploration Geophysics I**
Department of Earth, Ocean and Atmospheric Sciences,
University of British Columbia
- 2017–2019 **Statistical Mechanics**
Department of Geophysics and Astronomy,
Facultad de Ciencias Exactas, Físicas y Naturales,
Universidad Nacional de San Juan
- 2017–2019 **Physics I**
Department of Geophysics and Astronomy,
Facultad de Ciencias Exactas, Físicas y Naturales,
Universidad Nacional de San Juan

- 2013–2015 **Physics I**
 Physics Department,
 Escuela de Ciencias Exactas y Naturales
 Facultad de Ciencias Exactas, Ingeniería y Agrimensura,
 Universidad Nacional de Rosario
- 2013–2015 **Introduction to Science**
 Physics Department,
 Escuela de Ciencias Exactas y Naturales
 Facultad de Ciencias Exactas, Ingeniería y Agrimensura,
 Universidad Nacional de Rosario

WORKSHOPS AND TUTORIALS

- 2024 **Tutorial: Introduction to SimPEG at Agrogeo24**
 Devin C. Cowan, [Santiago Soler](#) and [Lindsey Heagy](#)
 Instructor
<https://doi.org/10.62329/PMXX7343>
- 2023 **BIRS 2-day Workshop: *Open-Source Tools to Enable Geophysical Data Processing and Inversion***
[Santiago Soler](#), [Lindsey Heagy](#), [Craig Miller](#) and [Leonardo Uieda](#)
 Workshop organizer
<https://birs-2023.softwareunderground.org>
- 2021 **Tutorial: Processing gravity and magnetic data with Harmonica**
[Santiago Soler](#), [Andrea Balza Morales](#) and [Agustina Pesce](#)
 Transform21, Software Underground
<https://youtu.be/0bxZcCAr6bw>
- 2020 **Introducción a Python para Científicxs**
 Universidad Nacional de San Juan, San Juan, Argentina
https://youtu.be/LS_e9gqTM2s
- 2020 **Tutorial: From scattered data to gridded products using Verde**
[Leonardo Uieda](#) and [Santiago Soler](#)
 Transform 2020, Software Underground
<https://youtu.be/-xZdNdvzm3E>
- 2017 **Taller Introductorio a LaTeX: Cómo producir documentos de calidad**
[Agustina Pesce](#) and [Santiago Soler](#)
 Universidad Nacional de San Juan, San Juan, Argentina

Supervision

- 2025 **Kelly Baker**, *University of British Columbia*
Summer Research Project.
Open-source implementation of gravity and magnetic forward model of ellipsoids.
Co-advised with [Lindsey Heagy](#).

Publications

PEER-REVIEWED SCIENTIFIC ARTICLES

- 2022 **New procedure to estimate plasma parameters through the q-Weibull distribution by using a Langmuir probe in a cold plasma**, *Plasma Research Express*
F.J. Gonzalez, J.I. Gonzalez, [Santiago Soler](#), C.E. Repetto, B.J. Gómez and D.B. Berdichevsky
doi: [10.1088/2516-1067/ac4f35](https://doi.org/10.1088/2516-1067/ac4f35)
- 2021 **Gradient-boosted equivalent sources**, *Geophysical Journal International*
[Santiago Soler](#) and [Leonardo Uieda](#)
doi: [10.1093/gji/ggab297](https://doi.org/10.1093/gji/ggab297)
- 2020 **Pooch: A friend to fetch your data files**, *Journal of Open-Source Software*
[Leonardo Uieda](#), [Santiago Soler](#), Rémi Rampin, Hugo vanKemenade, Matthew Turk, Daniel Shapero, Anderson Banihirwe and John Leeman
doi: [10.21105/joss.01943](https://doi.org/10.21105/joss.01943)
- 2019 **Gravitational field calculation in spherical coordinates using variable densities in depth**, *Geophysical Journal International*
[Santiago Soler](#), [Agustina Pesce](#), Mario Giménez and [Leonardo Uieda](#)
doi: [10.1093/gji/ggz277](https://doi.org/10.1093/gji/ggz277)
- 2018 **Transient plate contraction between two simultaneous slab windows: Insights from Paleogene tectonics of the Patagonian Andes**, *Journal of Geodynamics*
Guido M. Gianni, [Agustina Pesce](#) and [Santiago Soler](#)
doi: [10.1016/j.jog.2018.07.008](https://doi.org/10.1016/j.jog.2018.07.008)
- 2017 **Analysis of the Illapel MW=8.3 thrust earthquake rupture zone using GOCE-derived gradients**, *Pure and Applied Geophysics*
Orlando Álvarez, [Agustina Pesce](#), Mario Giménez, Andrés Folguera, [Santiago Soler](#) and Wenjin Chen
doi: [10.1007/s00024-016-1376-y](https://doi.org/10.1007/s00024-016-1376-y)

- 2017 **Effective elastic thickness in the Central Andes. Correlation to orogenic deformation styles and lower crust high-gravity anomaly**, *Journal of South American Earth Sciences*
 Héctor P.A. García, Guido M. Gianni, Marianela N. Lupari, Marcos A. Sánchez, [Santiago Soler](#), Francisco Ruiz and Federico G. Lince Klinger
 doi: [10.1016/j.jsames.2017.11.021](https://doi.org/10.1016/j.jsames.2017.11.021)

NON-REVIEWED ARTICLES

- 2023 **Fatiando a Terra: a journey into open-source software for Geophysics**, [EGU Geodynamics Blog](#)
[Santiago Soler](#)
<https://blogs.egu.eu/divisions/gd/2023/07/19/fatiando-a-terra>

Presentations

INVITED SPEAKER

- 2024 **Modernizing gravity processing: use the gravity disturbance**, *University of Houston*
 Invited by Dr. Jiajia Sun
 doi: [10.6084/m9.figshare.25676571](https://doi.org/10.6084/m9.figshare.25676571)
- 2023 **Processing potential field data with Fatiando a Terra**, *AGU Fall Meeting 2023*
 Abstract: <https://agu.confex.com/agu/fm23/meetingapp.cgi/Paper/1443596>
 doi: [10.6084/m9.figshare.24687573](https://doi.org/10.6084/m9.figshare.24687573)
- 2023 **Fatiando a Terra: software libre para geofísica**, *Instituto de Geociencias Básicas, Aplicadas y Ambientales (IGeBA), Universidad de Buenos Aires*
 Slides: <https://www.santisoler.com/2023-fatiando-igeba/>
- 2022 **Fatiando a Terra: Open-source tools for geophysics**, *Geophysical Inversion Facility, Department of Earth, Ocean and Atmospheric Sciences, University of British Columbia*
 Slides: <https://www.santisoler.com/2022-ubc-fatiando>
- 2022 **Empowering science with open-source software**, *Instituto de Astronomia, Geofísica e Ciências Atmosféricas, Universidade de São Paulo*
 Slides: <https://www.santisoler.com/iag-usp-2022>
- 2021 **Fatiando a Terra: Open-source tools for geophysics**, *Geophysical Society of Houston*
[Leonardo Uieda](#), [Santiago Soler](#) and [Agustina Pesce](#)
 Slides: <https://www.fatiando.org/2021-gsh>

CONFERENCE PROCEEDINGS

- 2026 **On the redesign of SimPEG's inversion framework**, *SEG: Critical Minerals Forum 2026*
- 2026 **A new inversion framework for SimPEG**, *EMIW 2026*
Poster: [10.5281/zenodo.21650406](https://zenodo.org/record/21650406)
doi: [10.5281/zenodo.21650406](https://zenodo.org/record/21650406)
- 2026 **Open-source gravity and magnetic forward model of ellipsoids**, *EGU 2026, General Assembly*
Abstract: <https://github.com/santisoler/emiw2026>
doi: [10.5194/egusphere-egu26-8366](https://zenodo.org/record/21650406)
- 2025 **3D geophysical inversions to characterize carbon sequestration potential of ultramafic rocks**, *EGU 2025, General Assembly*
Poster: [10.6084/m9.figshare.28811126](https://figshare.com/record/item?identifiers=figshare%3A28811126)
doi: [10.5194/egusphere-egu25-13551](https://zenodo.org/record/21650406)
- 2024 **Pooch: A friend to fetch your data files**, *SciPy 2024*
Talk: <https://www.youtube.com/watch?v=91q1Hadl9uI>
Slides: [10.6084/m9.figshare.26312236.v1](https://figshare.com/record/item?identifiers=figshare%3A26312236)
- 2023 **Pooch: A friend to fetch your data files**, *PyCascades 2023*
Lightning talk: <https://www.youtube.com/watch?v=KvxBc4xUMyg>
Slides: [10.6084/m9.figshare.22300822](https://figshare.com/record/item?identifiers=figshare%3A22300822)
- 2023 **Fatiando a Terra: Open-source tools for geophysics**, *Canadian Exploration Geophysical Society (KEGS)*
[Santiago Soler](#) and [Lindsey Heagy](#)
Slides: [10.6084/m9.figshare.22151357](https://figshare.com/record/item?identifiers=figshare%3A22151357)
- 2021 **Gradient-boosted equivalent sources for gridding large gravity and magnetic datasets**, *EGU21 General Assembly*
[Santiago Soler](#) and [Leonardo Uieda](#)
Talk: [10.6084/m9.figshare.14515188](https://figshare.com/record/item?identifiers=figshare%3A14515188) Slide: [10.6084/m9.figshare.14461792](https://figshare.com/record/item?identifiers=figshare%3A14461792)
doi: [10.5194/egusphere-egu21-1276](https://zenodo.org/record/21650406)
- 2020 **How to grid gravmag data with Harmonica**, *Transform 2020, Software Underground*
Lightning talk: <https://youtu.be/NtBDf7d7mwM?t=4245>
- 2020 **A better strategy for interpolating gravity and magnetic data**, *EGU2020, General Assembly*
[Santiago Soler](#) and [Leonardo Uieda](#)
doi: [10.6084/m9.figshare.12217973](https://figshare.com/record/item?identifiers=figshare%3A12217973)

- 2020 **Evaluating the accuracy of equivalent-source predictions using cross-validation**, *EGU2020, General Assembly*
[Santiago Soler](#) and [Leonardo Uieda](#)
doi: [10.5194/egusphere-egu2020-15729](https://doi.org/10.5194/egusphere-egu2020-15729)
- 2019 **Experiencias en el desarrollo de Fatiando a Terra**, *Taller Argentino de Computación Científica*
doi: [10.6084/m9.figshare.10013006](https://doi.org/10.6084/m9.figshare.10013006)
- 2019 **Gravitational fields of tesseroids with variable density**, *LAPIS 2019*
[Santiago Soler](#), [Agustina Pesce](#), Mario Giménez and [Leonardo Uieda](#)
doi: [10.6084/m9.figshare.8242439](https://doi.org/10.6084/m9.figshare.8242439)
- 2017 **Tesseroides con densidad variable: modelo directo con software libre**, *I Congreso Binacional de Investigación Científica y V Encuentro de Jóvenes Investigadores*
[Santiago Soler](#), [Agustina Pesce](#), [Leonardo Uieda](#) and Mario Giménez
- 2017 **Magnetic characterization of the Loncopué trough, Argentina**, *XX Congreso Geológico Argentino*
[Agustina Pesce](#), Mario Giménez, Andrés Folguera, Guido M. Gianni and [Santiago Soler](#)
- 2017 **Magnetic characterization of the Loncopué trough, Argentina**, *XX Congreso Geológico Argentino*
[Agustina Pesce](#), Mario Giménez, Andrés Folguera, Guido M. Gianni and [Santiago Soler](#)
- 2017 **Anomalías gravimétricas y magnéticas corticales del sur de la provincia volcánica de La Payenia, asociadas al tearing de la Placa de Nazca y anomalías mantélicas**, *XX Congreso Geológico Argentino*
Ana Astort, Bruno Colavitto, Lucía Sagripanti, Héctor P.A. García, [Santiago Soler](#), Francisco Ruiz and Andrés Folguera
- 2016 **Análisis flexural de la cuenca cretácico-paleógena del noroeste argentino. La subcuenca Lomas de Olmedo: zona de transición entre dos mecanismos de deformación distintos**, *Primer Simposio de Tectónica Sudamericana*
Héctor P.A. García, [Santiago Soler](#), Guido M. Gianni and Francisco Ruiz
- 2016 **Crustal Magmatic Anomalies from the Southern Payenia Volcanic Plateau, Associated with The Nazca Plate Tearing and Plume Head from Gravimetric and Magnetic Data**, *Primer Simposio de Tectónica Sudamericana*
Ana Astort, Francisco Ruiz, Héctor P.A. García, [Santiago Soler](#), Andrés Echaurren and Andrés Folguera

Reviewer

[Geophysics](#)

[Journal of Open Research Software](#)

[Journal of Applied Geophysics](#)

[Journal of Open-Source Software](#)

Proposals for [BIRS](#) workshops

Technical Skills

Programming Python, Rust, bash, FORTRAN, C

Scientific Python NumPy, Matplotlib, Pandas, Xarray, Numba, Dask, SimPEG, PyGMT

Markup Markdown, LaTeX, HTML, RST

WebDev CSS, Bootstrap, PicoCSS, Static Site Generators (jekyll, zola, urubu), Jinja2 and Tera

DevOps GNU/Linux, Unix terminal, VIM, Neovim, git, GNU Make, SSH, Ansible, nginx, ufw

Graphic Design Inkscape, GIMP, imagemagick, Darktable

Other tools Jupyter Notebooks, LibreOffice Suite, PGP, GitHub Actions, Nextcloud, maxima

Languages

Spanish Native

English Advanced

Italian Beginner